EcoPassEU: A Hybrid Blockchain Architecture for ESPR-Compliant Digital Product Passports

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1 Introduction

Billions of products enter the European market with sustainability labels every year, and consumers have little reliable means of verification for the information included, such as "low carbon" or "ethically sourced" [29]. A study commissioned by European Commission found more than half green environmental declarations without verifiable evidence to confirm them, eroding public confidence in environmental commitments [29].

To address the credibility gap, the Ecodesign for Sustainable Products Regulation (ESPR) now mandates Digital Product Passports (DPPs) -standardised records of a products' lifecycle- for most EU products [9].With the implementation of compliance in 2027, reliable DPP solutions have become a market urgent need.

Our product EcoPassEU is a blockchain-based DPP platform which allows manufacturers to record product information as permanent, publicly verifiable entries on the Ethereum blockchain. Unlike traditional databases, the immutability of blockchain records means that once the data is released, it cannot be altered retrospectively, eliminating the possibility of post-hoc falsification [7]. Any user can obtain a complete and encrypted product history by entering the product ID or scanning the QR code.

EcoPassEU serves four key stakeholders. It provides SME manufacturers with a cost-effective ESPR compliance channel through a web for batch data entry and automated QR code generation. At the same time, consumers, regulators and retailers can also get open and wallet-free access. Just search for the product ID to immediately obtain the blockchain-verified product passport and supply chain map, enabling the independent verification of environmental claims without prior technical expertise.

The value of EcoPassEU is threefold. It gives manufacturers a practical pathway to meet ESPR deadlines, reducing the risk of market exclusion. The platform also combats greenwashing: securing product records via cryptographic hashing replaces brand self-reporting with independent, third-party verification. Finally, it supports the circular economy by equipping recyclers with precise material data necessary to recover resources at end-of-life.

2 Background

2.1 Digital Product Passports and the Regulatory Context

A Digital Product Passport (DPP) is a structured digital record encompassing lifecycle data—such as material composition, provenance, and end-of-life guidance—designed to inform stakeholders and support more sustainable design choices, efficient resource use, and the transition towards circular business models [1,3,6,19].

The regulatory foundation for DPPs comes from the European Green Deal and Circular Economy Action Plan [15,16]. Building on these two frameworks, the Ecodesign for Sustainable Products Regulation (ESPR) has come into force from 18 July 2024 for products in the scope of the regulation, mandating DPPs for almost all physical goods sold in the EU, regardless of where the goods are sold [9].

2.2 The Problem: Trust in Sustainability Claims

Despite regulatory momentum, most DPP implementations rely on centralised, proprietary databases [27]. This architecture encourages selective concealment of data, hindering aftermarket participants such as independent repairers [10]. Managing large volumes of lifecycle data centrally also raises important concerns regarding its storage efficiency, data integrity, and vulnerability to breaches [33]. As such, downstream actors, including importers, recyclers and end consumers, are left with no reliable means to independently verify product information.

Greenwashing compounds this problem. Broadly understood as false, vague, or unsubstantiated environmental claims [22], it is widespread: a 2020–2021 EU sweep of 344 online claims found that nearly half were potentially deceptive—often substituting verifiable details with vague terms like “conscious”—and 59% lacked accessible evidence [29]. A separate EU website screening found that 42% of sustainability claims were misleading enough to constitute improper commercial behaviour [29].

Blockchain technology offers a structural solution to this issue. Because records are replicated across a distributed network, no single party can tamper with the data without detection [7,25]. It is important to note that while blockchain cannot guarantee that the data is initially physically true (the “oracle problem”) [11], it can ensure strict tamper-evidence, meaning that upon the commitment of a manufacturer to a sustainability claim, they are held publicly accountable for that claim [35].

2.3 Related Work and Existing Platforms

Blockchain-based product traceability uses its immutability and decentralised verification to improve supply chain transparency [7]. Blockchain-based product traceability emerged as a research topic around 2015–2016, with early work concentrated in food and agriculture before spreading to pharmaceuticals, electronics, and textiles [12,30]. More recently, research has shifted from feasibility towards scale and efficiency: for instance, Walmart's use of IBM Food Trust — built on Hyperledger Fabric — to reduce food-item tracing time from six days to under three seconds [12] .

In the commercial field, Circularise is among the closest comparable platforms, offering DPPs with selective disclosure of sustainability data across supply chains, with applications spanning the automotive, battery, and textile sectors [8]. VeChain similarly deploys blockchain and smart contracts on its VeChainThor platform to achieve supply chain transparency, though it operates on a permissioned network with a restricted set of authority-approved validators [32]. Both platforms share a common limitation: reliance on proprietary infrastructure or permissioned networks introduces vendor dependency and reduces interoperability [4,34].

EcoPassEU takes a different approach. Rather than relying on a proprietary database, it is built on the public Ethereum blockchain and IPFS, with all rules encoded in open smart contracts. Product data is stored on IPFS, whilst only a small set of cryptographic identifiers is written on-chain, keeping costs low. The batch registration further reduces the cost of individual products and makes the platform more attractive to SMEs. Moreover, as an open source project, it does not have the risk of vendor lock-in.

2.4 Business Viability

EcoPassEU's revenue model rests on two foundations. One is public funding: the EU's CIRPASS-2 Project (2024—2027) explicitly subsidise DPP deployment across textiles, electronics, and construction, with dedicated support for SMEs adopting "DPP-as-a-Service" models [17]. As an open source platform, EcoPassEU is fully capable of benefitting from such grants.

The other is subscription revenue. ESPR compliance is legally mandatory for any business selling physical goods in the EU, forming a captive market independent of voluntary commitments [9]. The green technology and sustainability market is projected to grow from USD $25.47 billion in 2025 to USD$73.90 billion in 2030 [24]. EcoPassEU charges a graded subscription fee based on passport volume, with optional premium tiers for API integration and compliance reporting. The platform is built on public Ethereum, supporting payment with ETH or other ERC-20 tokens, providing international users a lower-friction settlement option and removing the need for traditional banking intermediaries.

3 Design Documentation

3.1 Design Philosophy and Architectural Choices

EcoPassEU is a digital product passport (DPP) designed to provide an ESPR-compliant digital product passport and contains independently verifiable sustainability data. The design follows three design principles, each of which is aimed at specific product needs.

First, the hybrid storage model is adopted: the complete product data is fixed on IPFS through Pinata [5], and only the content identifier (CID), `keccak256` integrity hash, issuer address and timestamp [35] are recorded on the chain, thus reducing gas costs. Second, the system is privacy-aware and future-proof. While full data is currently stored off-chain on public network IPFS, this architecture sets the necessary foundation for data privacy. In future versions, manufacturers will also be able to easily encrypt commercially sensitive product data before uploading it. The on-chain hash will continue to function as an integrity anchor, proving the data has not been altered without exposing the full contents to the public ledger. Third, all read queries use public RPC endpoints, allowing consumers and regulators to view any passport without holding wallets or paying gas fees.

3.2 System Architecture

The user layer comprises 3 separate actor types: manufacturers, who connect to the platform via MetaMask and sign/submit transactions, and consumers/ regulators, who can query product records without holding a wallet. This is a non-exhaustive overview, as both actors must pass through the application layer — a React/Vite frontend deployed on the cloud platform Vercel that makes use of Ethers.js v6 for contract calls — to perform document uploads to IPFS, contract calls to the Sepolia network, and local integrity checks. These actor types, with differing requirements, are served by EcoPassEU (See Figure 1).

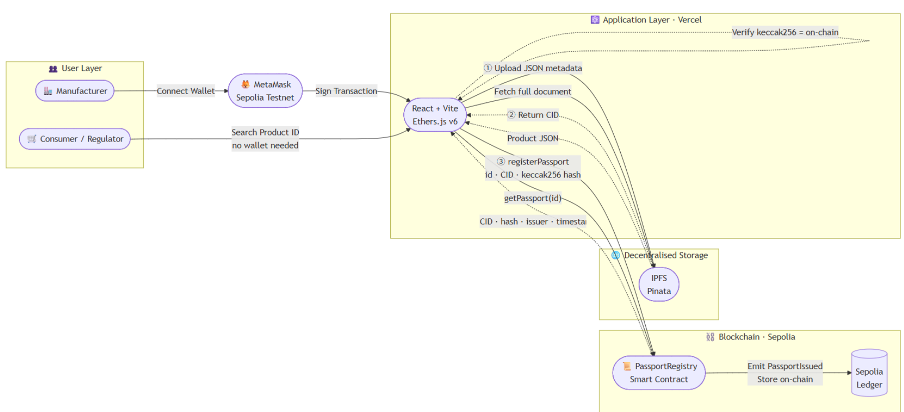

Fig. 1. Overall system architecture.

3.3 Use Cases and Workflow

EcoPassEU serves three actor types with differing technical needs, as illustrated in Figure 2.

Wallet connection via MetaMask on the Sepolia testnet (UC-01) is a common premise for all manufacturers and administrators, as each on-chain write must be signed by a wallet key pair and cryptographically linked to an accountable issuer [20,21]. To register a product, the manufacturer fills in a structured registration form (UC-02), including product details, material composition, and supply chain sources. Upon submission, the frontend serialises the form data to JSON and uploads it to IPFS via Pinata (UC-03) to receive a CID.

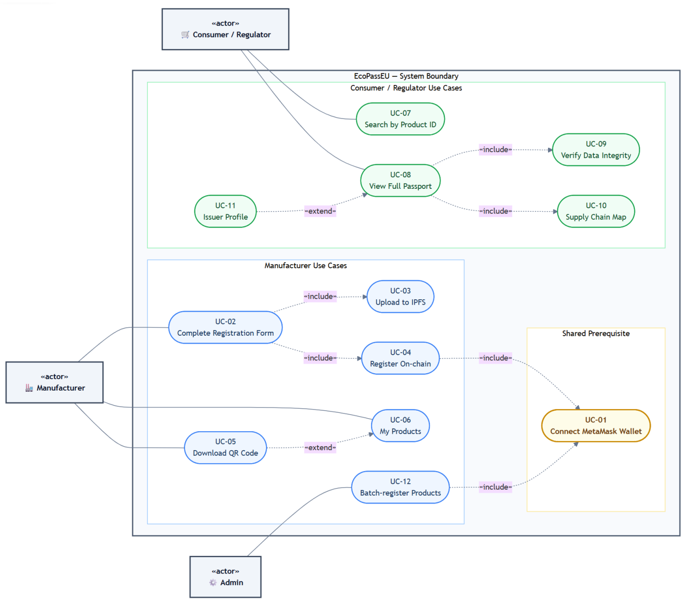

Fig. 2. EcoPassEU use case diagram.

Next, the system calculates the `keccak256` hash of the JSON locally and passes it to the `registerPassport()` function on the smart contract alongside the CID and product ID (UC-04). After the user confirms the transaction in MetaMask, a `PassportIssued` event is emitted, and a QR code with an Etherscan link is provided for users to download (UC-05), which also accesses the product list view (UC-06). The admin role requires the same wallet connection but adds batch registration capabilities (UC-12) to reduce the unit gas cost for high-volume producers.

At the read end, consumers and regulators can verify the product without a wallet. The user submits the product ID (UC-07), and the frontend queries the contract through the public RPC endpoint and jumps to the complete product passport page (UC-08). If the product passport exists, the system will obtain the complete JSON data from IPFS and recalculate its hash value on the client: if it matches, the green integrity badge will be displayed; if it does not match, a warning (UC-09) will be displayed. The product passport page will also display an interactive supply chain map (UC-10) and an optional issuer profile (UC-11). This separation between the write-gated and read-open flows supports the design goal of providing transparent information without setting financial or technical barriers to end users.

3.4 Sequence and State Diagrams

Figure 3 outlines the technical details behind the core application flows, focussing on the strict execution order required for successful registration. Specifically, the frontend must first obtain a valid CID from IPFS before prompting the wallet signature and triggering the smart contract. This ensures that the on-chain pointer never leads to data loss. Meanwhile, the verification flow demonstrates how the system relies entirely on the client logic to compare the acquired IPFS documents against the on-chain hash, ensuring trustless validation.

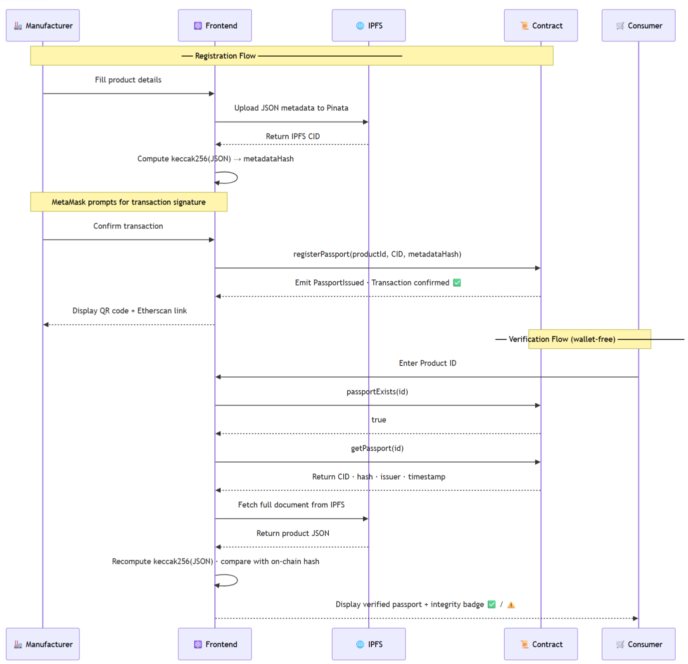

Fig. 3. UML sequence diagram.

Figure 4 illustrates a state machine diagram that describes the lifecycle that a digital passport undergoes. At the beginning of the process, physical products were always entered as `Unregistered`. The status of the digital document only changes to `IPFSStored` after the data is successfully uploaded to Pinata. Once the user signs the document, the status of the document changes to `TxPending`. After the block is mined on the Sepolia network, the digital document becomes immutable and is then marked as `Registered`. During the final verification step, the client hash comparison will mark the record as either `Verified` when the hash is similar or `Compromised` when the data has been altered.

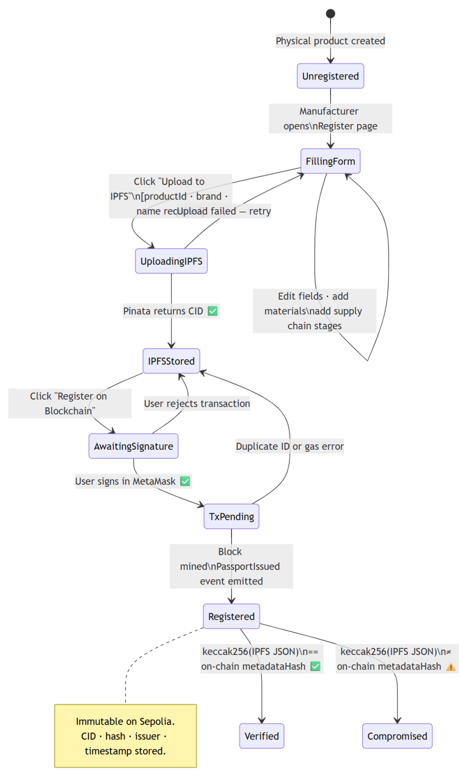

Fig. 4. UML state machine diagram.

3.5 User Interface Design

To reflect the environmental theme, the search page adopts a dark green colour scheme (#2d6a4f, #1b4332). Figure 5 shows that the search page is laid out with simple and consistent spacing. It presents a user-focused interface with a clear flow. Users can perform a search for a product by providing a product ID in the search box in the centre of the page or click on one of the pre-defined quick search buttons. No wallet connection is required. The interface begins with an empty state and upon a search is then replaced by a verified product result card. The card contains the crucial information on-chain data, such as the product issuer address, timestamp of registration on-chain, and the hash of product metadata. This allows users to instantly verify the authenticity of the product before accessing the full product passport.

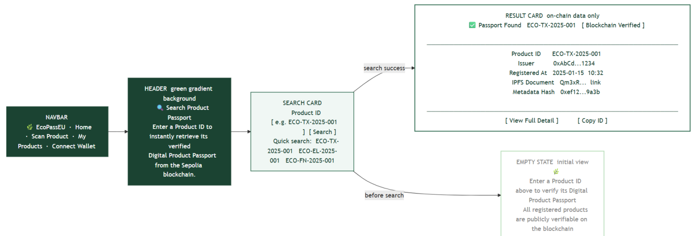

Fig. 5. Wireframe of the search interface.

Figure 6 displays the registration layout. To provide a clear representation of the user's current state in the registration process, a simple tracker is included on the page with different steps: Fill Details → Upload IPFS → Register Chain → Done. There is no hidden section; all fields within the product detail, materials, and supply chain forms are presented on one scrollable page, which avoids confusion where the user does not know which section has not yet been completed or submitted. If the page is reached without having MetaMask connected, an amber coloured banner will appear under the details section. By doing this, we can stop users from uploading their data if they have not logged in yet.

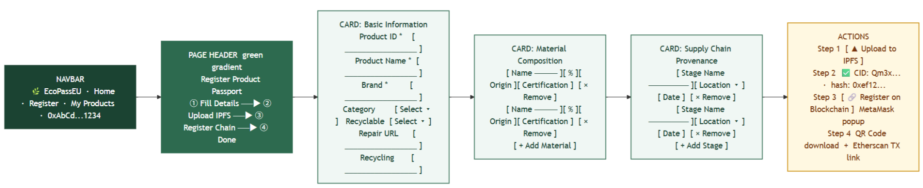

Fig. 6. Wireframe layout of the manufacturer registration form.

The result page is described in Figure 7, clearly differentiating information. The *Blockchain Record* card displays on-chain data (issuer, time, and hash), while other cards display IPFS details. It is easy to see whether the information is valid by only considering the integrity badge. It will be presented as a green checkmark if the check can be passed, or as an amber triangle if an error occurs. If a user does not install MetaMask, the page will automatically use public RPC to load information.

We've implemented a CSS Grid and a Flexbox to ensure our site works perfectly on any device. At 600,px, forms will switch to a single-column layout to adapt to the mobile device screen. We had to use a 768,px breakpoint for the menu and interactive map size, to make it easier for users on a tablet.

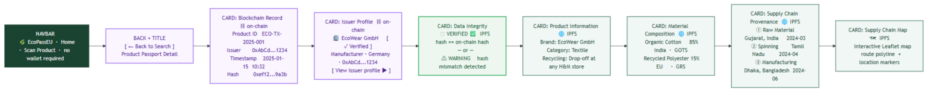

Fig. 7. Wireframe detailing the structure of the product verification page.

4 Project Execution

4.1 Core Code Analysis and Data Structures

EcoPassEU uses a hybrid architecture, separating on-chain and off-chain data to balance transparency with efficiency. The core data structure is divided across these two environments.

Off-chain, the complete lifecycle data—including category, brand, material composition, and supply chain provenance—is structured as JSON objects and pinned to IPFS via Pinata. On-chain, the `PassportRegistry.sol` smart contract uses a strictly typed `Passport` struct to store only the essential, immutable references:

Listing 1.1. Core Data Structure in PassportRegistry.sol

```
struct Passport {
        string  ipfsCID;       // off-chain
            document pointer
        bytes32 metadataHash;  // keccak256
            of JSON for tamper-evidence
        address issuer;
        uint256 timestamp;
        bool    exists;
}

// productId => Passport
mapping(string => Passport) private passports
    ;
```

This private mapping directly associates a unique product ID with its corresponding passport data for O(1) retrieval. Initially, we considered using ERC-1155 tokens to represent these passports; however, we opted for this lightweight registry approach. Storing only the IPFS CID, a cryptographic hash, the issuer address, and a block timestamp on-chain significantly reduces deployment complexity and gas costs per registration, whilst keeping the full product lifecycle data off-chain on IPFS.

Listing 1.2 outlines the core registration functions and reflects a number of optimisations for Solidity:

Listing 1.2. registerPassport function

```
function registerPassport(
string calldata productId,
string calldata ipfsCID,
bytes32 metadataHash
) external {
        require(!passports[productId].exists,
            "Passport already exists");
        passports[productId] = Passport({
                ipfsCID:      ipfsCID,
                metadataHash: metadataHash,
                issuer:       msg.sender,
                timestamp:    block.timestamp
                    ,
                exists:       true
        });
        emit PassportIssued(productId,
            ipfsCID, metadataHash,
        msg.sender, block.timestamp);
}
```

Passing string parameters as `calldata` instead of `memory` avoids redundant data copying and reduces gas consumption. The `external` visibility modifier further improves efficiency by allowing direct argument access from calldata, compared with a `public` declaration. What is more, binding `issuer` to `msg.sender`—a secure EVM primitive—provides unforgeable cryptographic attribution for every passport without requiring supplementary access controls.

As a supplement to the registration logic, the `PassportIssued` event will be issued after each successful execution:

Listing 1.3. PassportIssued Event Definition

```
event PassportIssued(
string  indexed productId,
string          ipfsCID,
bytes32         metadataHash,
address indexed issuer,
uint256         timestamp
);
```

By designating `productId` and `issuer` as `indexed`, EVM stores these parameters in the Bloom filter of the transaction receipt. This mechanism allows off-chain clients to efficiently filter logs by specific product or manufacturer without scanning the computing overhead of each block, which represents best-practice model of establishing queryable audit tracking in Ethereum DApps [35].

4.2 Testing Strategy

To validate the reliability of `PassportRegistry`, a unit-testing suite was implemented using the Hardhat environment alongside the Chai assertion library [?], following established practices for smart contract validation [18]. Listing 1.4 shows the shared test fixture; a fresh contract instance is deployed via `beforeEach`, ensuring full isolation between cases.

Listing 1.4. Hardhat test fixture – shared setup

```
beforeEach(async function () {
        [owner, addr1] = await hre.ethers.
            getSigners();
        const Factory =
        await hre.ethers.getContractFactory("
            PassportRegistry");
        contract = await Factory.deploy();
});
```

The most critical tests concern `registerPassport`, seen in Listing 1.5. The first case confirms that `msg.sender` is persisted as the issuer, which is the foundation of the platform's unforgeable attribution model. The second verifies that attempting to re-register an existing product ID triggers the `require` guard and reverts with the expected reason string, guaranteeing passport immutability.

Listing 1.5. Unit tests – registerPassport: issuer storage and duplicate guard

```
it("should store the correct issuer address",
     async function () {
        await contract.registerPassport(
            PRODUCT_ID, IPFS_CID, META_HASH);
        const [, , issuer] = await contract.
            getPassport(PRODUCT_ID);
        expect(issuer).to.equal(owner.address
            );
});

it("should revert on duplicate product ID",
    async function () {
        await contract.registerPassport(
            PRODUCT_ID, IPFS_CID, META_HASH);
        await expect(
        contract.registerPassport(PRODUCT_ID,
             IPFS_CID, META_HASH)
        ).to.be.revertedWith("Passport
            already exists");
});
```

4.3 Final UI Showcase

Figures 8 and 9 show the deployed interface of EcoPassEU.The registration portal guides manufacturers through a structured form (with step indicator), whilst the public verification page shows the passport details, integrity status and IPFS document link (without wallet requirement).

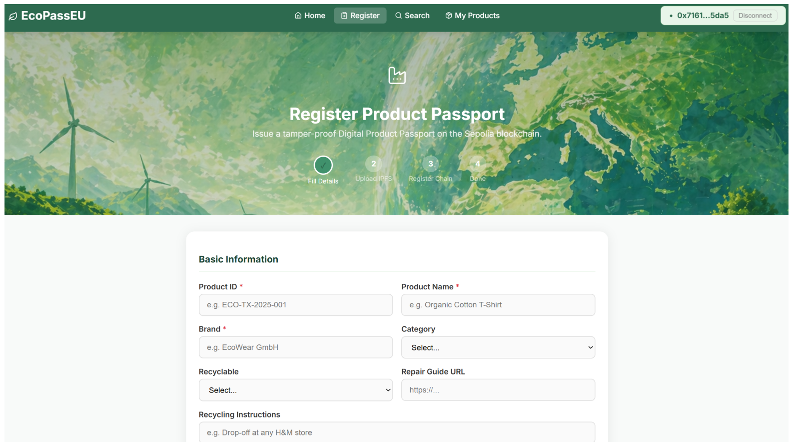

Fig. 8. Deployed EcoPassEU interface: manufacturer registration portal.

Fig. 9. Deployed EcoPassEU interface: public product verification page.

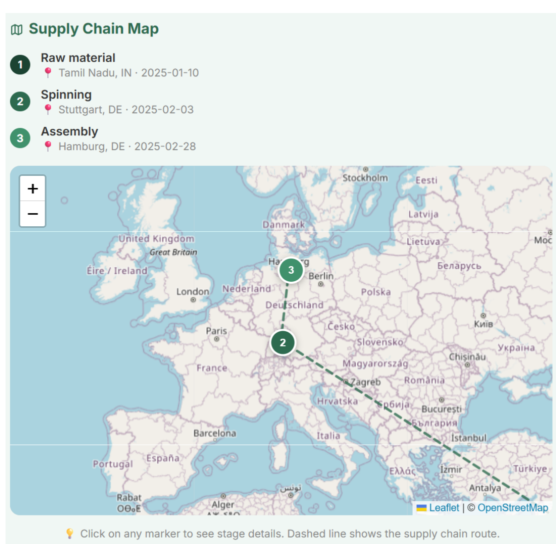

Fig. 10. Interactive supply chain map rendered on the public product verification page (UC-10).

Figure 10 illustrates the interactive supply chain map that displays every registered stage of the supply chain from a geographic perspective. By doing this, we enable consumers and regulators to follow the product's provenance, from the sourcing of raw materials to its final assembly.

4.4 User Manual

The application is designed to be intuitive for two distinct user groups.

For Manufacturers (Registration):

1. Connect your MetaMask wallet to the Sepolia testnet using the top navigation bar.
2. Navigate to the 'Register' page and complete the product detail forms, including material and supply chain data.
3. Click 'Upload to IPFS'. The system will generate a unique CID.
4. Click 'Register on Blockchain' and approve the transaction in the MetaMask pop-up.
5. Once confirmed, download the generated QR code to attach to your physical products.

For Consumers and Regulators (Verification):

1. No wallet connection needed. Open the EcoPassEU homepage.
2. Enter the Product ID into the central search bar, or scan the product's QR code.
3. Review the product's lifecycle data. There will be a green 'Blockchain Verified' badge, indicating that the IPFS data matches the immutable blockchain record.

5 Critical Evaluation

A per-transaction cost analysis of this DApp's operation reveals highly optimised gas efficiency. The transaction cost can be rigorously decomposed based on standard EVM opcode pricing [35]. Each registration transaction has a base cost of 21,000 gas, with an additional cost of approximately 20,000 gas per `SSTORE` opcode for each new storage slot written. By storing heavy string data (e.g., repair guides) off-chain on IPFS, the on-chain state changes can be kept extremely lightweight. Moreover, it's important to note that the `Passport` struct can write 5 fields, however, using `bytes32` type for `metadataHash` instead of a hex string halves the storage cost for this one field, since it fits perfectly into one 32-byte EVM storage word. The testnet deployment reveals that conducting a single registration transaction would use approximately 125,430 gas. However, it is critical to mention that there can be very high operational scalability provided by the `batchRegisterPassports` function. Since the contract iterates an array of products in one transaction, it effectively amortises the fixed 21,000-gas base cost across multiple registration transactions. Based on the test data, it has been shown that registering 10 products in bulk reduces the average per-product cost to roughly 92,100 gas. This in turn explains the 26% per-unit cost reduction, and gives the platform potential to be economically viable for SMEs.

In terms of security, the implementation has successfully reduced the risk of tampering and unauthorised modification, which is a central concern in Ethereum smart contract security [23,31]. In smart contracts, the explicit `require` check provides a strict tamperability detection guarantee by making the registered passport immutable. In addition, the stored metadata hash establishes a cryptographic commitment between the on-chain registry and the off-chain IPFS load. If the malicious actor tries to intercept and tamper with the IPFS document, the cryptographic hash of the tampered document will no longer match the immutable record saved on the chain, thus immediately invalidating the passport. Unlike existing solutions that rely on permissioned nodes or proprietary databases, EcoPassEU achieves enterprise-level privacy protection on a fully open network through the IPFS hash pairing mechanism.

The contract structurally avoids several common loopholes in Ethereum. The re-entry attack [23] is not applicable because the contract neither accepts nor transfers Ether, eliminating the external call path used by reentrancy attack. Integer overflow has been mitigated by Solidity 0.8.x's built-in checked arithmetic, which reverts automatically on overflow without requiring the SafeMath library [14]. The `private` visibility of `passports` mapping prevents direct external enumeration; all reading are controlled by the typed `getPassport()` and `passportExists()` view functions, which are declared as `view`, therefore cost zero gas when called off-chain.

Although the proposed EcoPassEU system presents strong advantages, it does have some limitations that arise from the decentralised architecture. While the on-chain verification process is extremely fast, obtaining the complete JSON from public IPFS gateways can have some longer latency compared to centralised servers. First and most importantly, even though the batch registration function is very affordable on a testnet, executing such loop-based operations directly on the Ethereum mainnet will get immensely pricey during congested networks, so the production version of EcoPassEU is intended primarily for deployment on Layer-2 networks (Polygon, for example, or Arbitrum). This approach will still allow EVM compatibility and security while also making near-zero transaction fees for long-term financial viability of SMEs [28].

6 Conclusion

In this work, we show that a proof-of-concept implementation (EcoPassEU) builds a workably tamper-proof infrastructure for Digital Product Passports according to the ESPR. By integrating Ethereum smart contracts [2] with the off-chain system, IPFS, we create independently verifiable sustainability records that do not compel end users to handle crypto wallets. Further, a combination of a `keccak256` hash pairing system [26] with Ethereum's Proof-of-Stake consensus [13] solves the fundamental integrity problem without the high carbon footprint that typically accompanies blockchains. Finally, the feature of batch registration results in a 26% reduction in gas costs, bringing much greater access for small and medium-sized enterprises (SMEs).

Nonetheless, the prototype has several limitations in practical use that must be overcome before launch to the public. First, using the testnet exclusively to test on the Sepolia testnet means that gas costs are untested in practice. Specifically, this is true for the loop-based `batchRegisterPassports` function, which in mainnet congestion could become prohibitively expensive. Another crucial problem lies in the off-chain storage, where, at the moment, it is purely dependent on Pinata. In the case of this single gateway being unavailable, the documents it links to will become inaccessible, leaving the passport cryptographically correct, but useless in use. As well, the whole system is completely open; there are no identity checks, allowing any wallet address to issue a passport, so providing no confidence that it came from the genuine manufacturer. Finally and most crucially, the strict immutability of the blockchain does not allow users to fix mistakes they make in good faith or to withdraw invalidated entries. This is a large issue for regulators, as, at present, there is also no upgradeable proxy pattern in the current smart contract design.

EcoPassEU will need a clear roadmap to transition from a proof-of-concept into a production-ready environment. As outlined earlier, the platform needs to be migrated into a Layer-2 solution to ensure costs are affordable when scaled. Additionally, the current proof-of-concept is not set up to have a formal identity verification mechanism. In future versions, W3C Decentralised Identifiers (DIDs) and Verifiable Credentials [36,37] should be included in the system. Not only will this link a manufacturer's wallet to a legally verified entity, but it solves the "oracle problem" without sacrificing corporate privacy.

Finally, addressing the strict immutability of the blockchain, an upgradeable and controlled proxy pattern should be considered. This would give regulators and manufacturers the ability to amend honest errors or update compliance requirements, without compromising historical passport records.

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A Appendix

This appendix includes the mandatory materials required for the FTGP summative assessment.

A.1 Smart Contract Address

– `https://sepolia.etherscan.io/address/0x0617635eA34a7835807EbC6D0A7aECC9de8E1Cf0`

A.2 Deployed Website Link

– `https://ftgp-2526-group1-11.vercel.app/home`

A.3 Transaction Hashes

– **Test Registration TX:** `https://sepolia.etherscan.io/tx/0xe371e32fe81af2087170b43d88234883b1577c12e6be347b54f7728d3ed64c46`
  Explanation: This transaction demonstrates a successful test execution of the `registerPassport` function. It securely anchors a test product's IPFS CID, cryptographic metadata hash, and the manufacturer's wallet address to the blockchain, subsequently emitting the `PassportIssued` event.
– **Deployment TX:** `https://sepolia.etherscan.io/tx/0x7925b3d7062d77f911b7614feeb15337ee7be0c31dc748ed216be717a2b433e8`
  Explanation: This transaction represents the initial successful deployment of the `PassportRegistry` smart contract onto the Sepolia testnet, establishing the immutable registry for EcoPassEU.

A.4 GitHub Repository Link

– `https://github.com/DeTaiDong/FTGP2526_Group1_11/tree/main`

A.5 Equity Shares

– Luxiao Cao: 1.01
– Akshansh Rajora: 1.01
– Fuyu Cao: 0.98
– Tianwei Yu: 0.97
– Detai Dong: 1.03

A.6 Screenshot of GitHub – Insights - Contributor

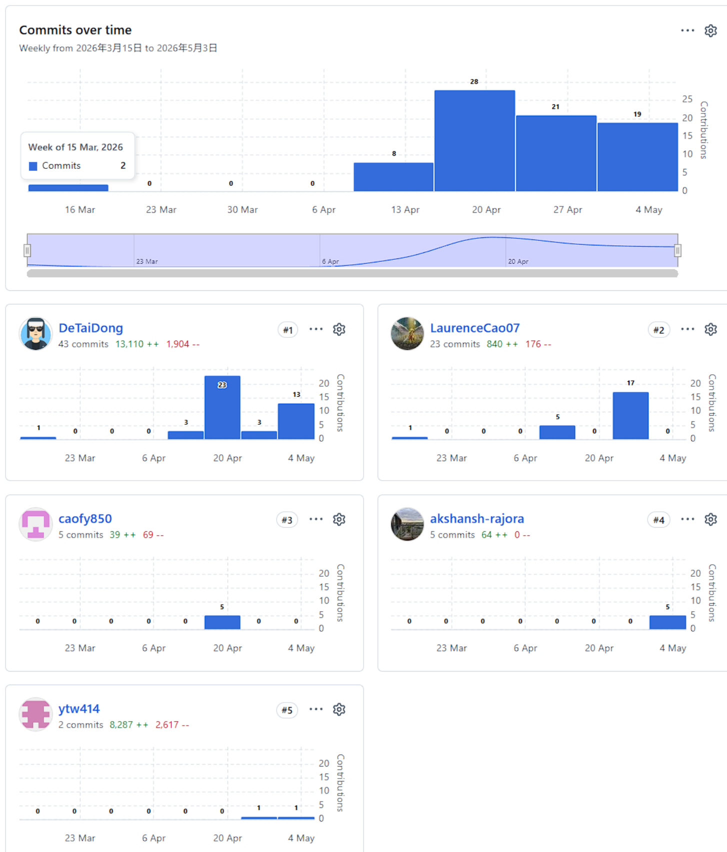

Fig. 11. Screenshot of GitHub – Insights - Contributor

A.7 Project Management and Collaboration

The team adopted a dual-platform approach to project management. We utilised a JIRA Scrum board for high-level sprint tracking and task status monitoring. Simultaneously, GitHub Issues were employed for detailed technical discussions and managing repository-specific milestones.

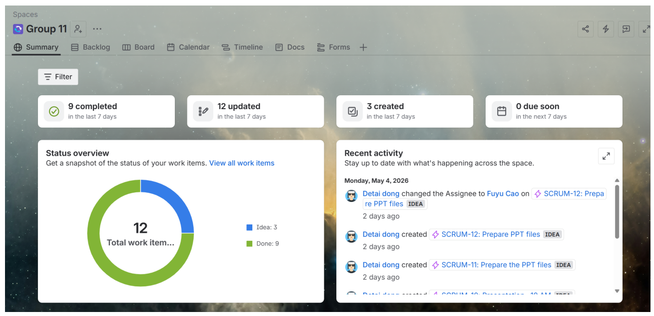

Fig. 12. JIRA Summary Board.

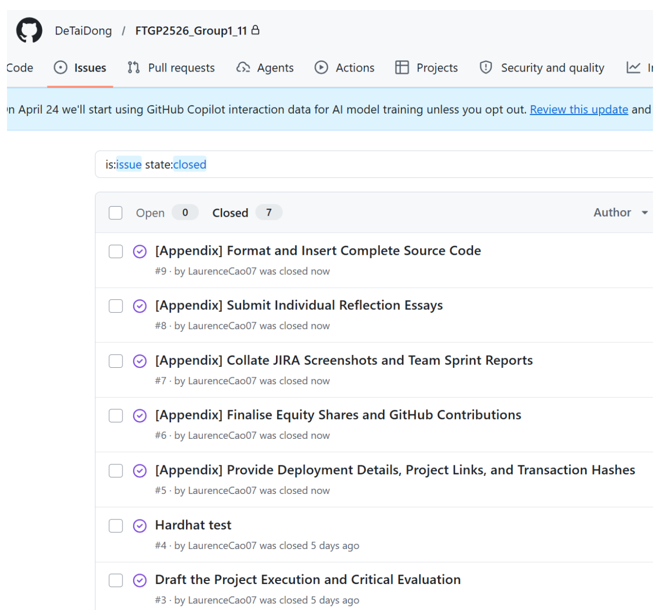

Fig. 13. GitHub Issue Tracking.

A.8 Individual Reflection Essays

Luxiao Cao Ruminating on my experience with the EcoPassEU group project, I served mainly as the lead technical writer and systems architect. My chief role included summarizing our team's technical development in a cohesive academic narrative, as well as describing the structural designs for our decentralised application (DApp). The project delivered a precious opportunity to blend theoretical blockchain concepts and practical, regulatory-driven software engineering approaches.

I was primarily involved in writing around 80% of the final project report. This involved undertaking thorough initial research into the current regulatory environment, particularly the Ecodesign for Sustainable Products Regulation (ESPR) and the widespread problem of corporate 'greenwashing' within the European Union. Through a rigorous literature review and carefully aggregating relevant academic references, I came to a deep understanding of Europe's strict commitment towards environmental sustainability and the circular economy. Synthesizing this contextual background was essential for articulating the intellectual and commercial motivations behind the EcoPassEU platform.

In addition to the academic writing, I led the visual systems modelling for the design documentation. I was responsible for carefully creating and producing the full suite of architectural diagrams, covering the overall system architecture, use case diagrams, UML sequence and state machine diagrams, and the user interface wireframes. The development of these models was extremely instructive; it forced me to systematically record the cross-cutting hybrid storage interactions between the React frontend, the IPFS off-chain storage and the Ethereum Sepolia testnet. Upon reflection, I realised that transforming the abstract interactions of the blockchain into clear, standardised visual workflows was necessary in aligning the team's development objectives and ensuring smooth integration.

In terms of the technical execution, I actively contributed to the Hardhat smart contract testing to ensure the robustness, immutability, and security of our Solidity code. Identifying vulnerabilities in the logic of our registry greatly deepened my knowledge of smart contract attack vectors and testing frameworks with a real-world case study. In addition, I contributed to the GitHub project documentation, making sure our repository was accessible, professionally presented, and well-structured.

Ultimately, this project has been an extremely transformative academic experience. Before this module, my understanding of distributed ledger technology and smart contracts was largely abstract. However, through the rigorous process of testing smart contracts, designing UML frameworks and communicating our technical findings, I have developed my understanding of blockchain ecosystems significantly. More generally, this project highlighted how new financial technologies could be effectively utilised to ensure transparency and accountability in global supply chains, ideal for and supporting modern European environmental pursuits.

Akshansh Rajora For the EcoPassEU group project, I focused on two main technical areas: the Solidity smart contract that serves as the foundation of the platform, and the IPFS integration that supports hybrid storage. Looking back, I can say it was one of the most technically challenging and rewarding tasks I have taken on during my degree. My first major task was designing and building 'PassportRegistry.sol'. The main challenge was not just to write a functioning contract but to create one that was secure, gas-efficient, and appropriate for real-world regulations. Instead of using the ERC-1155 token standard we initially considered, I chose a lighter registry pattern. This pattern involved a private mapping from product ID to a 'Passport' structure. This choice simplified deployment and cut per-registration gas costs since we only needed to store a CID, a 'bytes32' hash, an issuer address, and a timestamp on-chain rather than minting a full token. Every design decision had a direct financial impact, which helped me understand how software engineering on Ethereum differs from traditional backend development. The optimizations I made were very instructive. Declaring function parameters as 'calldata' instead of 'memory' prevents unnecessary data copying in the EVM. Using 'external' visibility instead of 'public' lowers call overhead for functions that are only accessed from outside the contract. Opting for 'bytes32' for the metadata hash instead of a hex string cut the storage cost for that field in half since it fits neatly into one 32-byte EVM storage slot. These details might seem minor, but together they added up to a significant 26 percent reduction in per-unit costs when combined with the 'batchRegisterPassports' function, which spreads the fixed 21,000-gas base cost across multiple products within a single transaction. I also devoted a lot of time to security. I examined the contract for common Ethereum vulnerabilities. Re-entrancy attacks were not a concern since the contract does not hold or transfer Ether. Integer overflow issues were automatically handled by Solidity 0.8.x's built-in checked arithmetic, eliminating the need for the SafeMath library. The 'require(!passports[productId].exists)' check ensures that every registered passport remains permanently unchanged, which aligns with ESPR compliance—meaning a manufacturer cannot simply overwrite a verified sustainability claim. Associating the issuer with 'msg.sender' guarantees unforgeable attribution without needing an additional access control layer. The IPFS integration required a different approach. I set up the workflow so that the frontend serializes product data into JSON, uploads it to IPFS via Pinata to get a CID, and computes the keccak256 hash of that JSON locally before writing both to the contract. This sequence is essential as the CID must exist before requesting a wallet signature to ensure that the on-chain pointer cannot lead to a lost-data situation. The hash-binding mechanism means that if an attacker modifies the IPFS document, the new hash will not match the unchangeable on-chain record, immediately indicating that the passport has been compromised. Working on this project showed me the limits of the architecture I created. The strict immutability makes correcting honest mistakes impossible without using an upgradeable proxy pattern. These are the issues I would focus on in a future version. Overall, this project provided me with real experience in production-level smart contract engineering, and I now have a much clearer understanding o the trade-offs between decentralization, cost, and accuracy.

Fuyu Cao During the FTGP group project, I mainly contributed to the frontend presentation, visual refinement, and part of the data preparation for our DApp, EcoPassEU. Our project focused on building a privacy-aware Digital Product Passport platform for sustainable consumer goods in Europe. Compared with some of my teammates who worked more on the smart contract logic or report writing, my role was more related to improving how the project was presented and how users would interact with it. I helped optimise the page design, adjust the layout of several pages, improve the homepage background and visual consistency, and design the project logo. In addition, I also processed part of the dataset so that it could be used more clearly in the frontend demonstration.

At the beginning of the project, I did not have a very strong background in frontend development, so some tasks seemed unfamiliar to me. For example, when I first looked at the project files, I was not very confident about how to modify the UI, replace images, or update components correctly. However, through this project I gradually became more comfortable with reading the code structure, locating the relevant files, and making direct changes to the interface. I learned how the React frontend was organised, how local changes could be previewed on the browser, and how visual elements such as logos, banners, and page sections could be integrated into the project. Although these may seem like small tasks compared with blockchain deployment, I realised that they are still very important because they directly affect the clarity, usability, and professionalism of the final product.

Another useful part of my work was helping with the dataset. The raw data was not in a form that could be directly used in the project, so I helped clean part of it and organise sample data that was easier for the frontend to display. This made me understand that in a real project, the gap between raw data and usable system data is often larger than expected. Even if the final interface looks simple, a lot of supporting work is needed behind it.

From a teamwork perspective, this project also taught me a lot. I learned that not every contribution has to be the "main technical core" in order to be valuable. In a group project, different members naturally take responsibility for different parts, and what matters is whether each person helps move the project forward. In my case, improving the UI, preparing data, and helping with presentation materials were practical contributions that supported the overall result. I also became more aware of the importance of communication, especially when multiple people are editing code, updating files, and trying to keep the project consistent.

Overall, this project was a meaningful experience for me. It not only improved my understanding of blockchain-based application development, but also gave me more confidence in contributing to a technical team project in my own way. I learned that building a complete product requires not only backend logic and smart contracts, but also design, data preparation, usability, and coordination.

This experience helped me better understand how a group project comes together as a whole.

Tianwei Yu Reflecting on the EcoPassEU project, I am incredibly proud of what our team accomplished. Our goal was to build a Digital Product Passport (DPP) platform to help European consumers verify if products are truly sustainable, fighting back against deceptive "greenwashing". From the begining, I actively participated in our group discussions, brainstorming ideas and helping shape the direction of our hybrid blockchain solution. This vibrant, collaborative energy was the driving force behind our success.

To build a reliable system, we first needed solid data. I teamed up with Fuyu to handle the critical task of data collection and cleaning. We worked carefully to ensure that the product lifecycle and supply chain data feeding into our system was accurate, well-structured, and ready for deployment. But a great backend also needs a great frontend. I wanted our platform to look as good as it works, so I collaborated closely with Fuyu again to beautify the website's user interface. We polished the page backgrounds, refined the text descriptions to make them highly readable, and carefully selected intuitive icons. By focusing on a clean, nature-inspired green theme, we ensured that anyone—whether a factory manager or an everyday shopper—could easily use our platform without needing any prior technical expertise[cite: 1].

On the technical side, I took charge of making sure our blockchain logic was bulletproof. A decentralized app is only as good as its security, so I conducted rigorous Hardhat smart contract testing. I wrote detailed test cases to guarantee that our 'registerPassport' function worked flawlessly, correctly recording the manufacturer's address and preventing duplicate product registrations. Beyond testing, I also conducted the Etherscan contract verification. This was a vital step for transparency, as it allowed anyone to publicly view and verify our smart contract code on the network. It felt incredibly rewarding to bridge the gap between our secure on-chain Ethereum records and our off-chain IPFS data storage.

Finally, bringing a complex project to life requires excellent documentation. I worked hand-in-hand with Fuyu, Luxiao, and the rest of the team to draft, supplement, and heavily refine our final project report. We combined our insights to ensure that every architectural choice we made was explained clearly and professionally.

Overall, EcoPassEU was much more than just a coding assignment; it was a chance to use cutting-edge technology to create real-world environmental impact. Whether I was cleaning data, polishing the visual UI, testing smart contracts, or writing the report, I learned how to build robust, user-centric decentralized applications. This journey has deeply strengthened both my software engineering skills and my appreciation for great teamwork.

Detai Dong In this project, I developed the GitHub repository and performed full-stack front-end development work. I laid out the overall framework comprising of the Home, Register, and Product Detail pages and defined the layout, visual theme, and interactive feedback components (e.g., the navigation bar, search bar and product list cards) used for all pages. The primary background artwork was provided by Fu Yu Cao. I developed the wallet connection component and designed the Landing page, connected to the smart contract developed by Akshansh, and — with his help — finished implementing the whole pipeline of uploading product metadata to IPFS and attaching the obtained CID to the blockchain. Subsequent to the DApp testing conducted by Tianwei Yu, I proceeded with the implementation of product search and the detail page, covering: issuer profile viewing and editing, per-product QR code generation, IPFS-based data integrity verification, supply chain visualisation on the world map, and composition of product materials. Moving into the future, I intend to improve on the functionality and layout of the Product Detail page constantly, always looking for ways to further enhance it.

This was my first time working with React and JSX. As compared to previous development experience, I noticed that component-based development allows for much higher modularity and reusability than plain HTML (which notably helped iteration speed when project scope grew). The most difficult component for me to comprehend was how the front end interacts with a smart contract — querying on-chain events, transaction data parsing and the asynchronous nature of blockchain responses all required careful consideration. Beyond this, debugging infuriating errors was equally challenging, but troubleshooting each of these errors contributed meaningfully to my development experience. Witnessing each feature working end-to-end provided tremendous motivation to me; I often would code myself for innumerable hours without even noticing the passage of time, which can certainly be seen in the bursts of focused activity carried out in GitHub. The motivation this project provided me with helped me to regain my passion for development and allowed me to accumulate hands-on Web3 experience which theoretical education could not provide.

Finally, I would like to express my heartfelt appreciation to the lecturers and supervisors of SEMTM0029_2025_TB-2, as well as my team members, for providing me with the opportunity to perform full-stack development tasks—an experience that significantly helped me strengthen both my technical skills and my understanding of Web3.

A.9 Team Sprint Reports

FTGP Group 11: Sprint Report 1

Luxiao Cao[1], Akshansh Rajora[1], Fuyu Cao[1], Tianwei Yu[1], and Detai Dong[1]

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1 Team Members

- Luxiao Cao (ne25392)
- Akshansh Rajora (tk25240)
- Fuyu Cao (ld25515)
- Tianwei Yu (np25024)
- Detai Dong (mz25325)

2 Product Vision Summary

EcoPassEU is a privacy-preserving Digital Product Passport (DPP) DApp designed for sustainable consumer goods in Europe, with an initial focus on textiles and small electronics. By scanning a QR code or NFC tag, users access verified product provenance, material composition, and repair/recycling guidance.

The project utilises a hybrid architecture: sensitive documents are stored off-chain, whilst cryptographic hashes, timestamps, and ownership proofs are maintained on-chain. This approach guarantees traceability and regulatory compliance without compromising commercial confidentiality, delivering a practical prototype for SMEs, consumers, and regulators.

3 Product Backlog

The current product backlog consists of the following foundational tasks:

- Define MVP scope and core user stories.
- Design the hybrid system architecture (on-chain and off-chain components).
- Set up the GitHub repository, branching strategy, and the Jira board.
- Design the product passport data structure for textiles and electronics.
- Build the initial Solidity smart contract skeleton for record creation.
- Explore and test IPFS-based off-chain document storage.
- Create the React frontend structure and core pages.
- Prototype a QR-code-based passport lookup flow.
- Prepare project documentation, sprint reports, and presentation materials.
- Review GDPR compliance and data immutability constraints.

Table 1. Sprint 1 Task Allocation and Details

Task	**Description**	**Owner(s)**	**Due Date**
Confirm MVP scope & user stories	Clarify prototype inclusions/exclusions. Identify manufacturer, consumer, and regulator roles.	Luxiao Cao	17th Mar
Set up GitHub & Jira board	Create shared repository, define branching/issues, and generate Sprint 1 backlog on Jira.	Detai Dong	17th Mar
Design system architecture	Produce diagram showing blockchain, IPFS, frontend, and QR/NFC. Agree on on/off-chain data split.	Akshansh Rajora	18th Mar
Define product data model	Decide core fields (ID, materials, provenance, repair/recycling). Prepare sample test records.	Tianwei Yu	19th Mar
Set up smart contract environment	Initialise Solidity/Hardhat project. Create skeleton for batch registration and metadata hash storage.	Fuyu Cao	20th Mar
Test IPFS off-chain storage	Upload sample JSON to IPFS, retrieve CID/hash, and verify smart contract linkage.	Akshansh Rajora	21st Mar
Create frontend scaffold	Set up React structure. Build Home, Passport View, and Issuer Upload Form placeholder pages.	Tianwei Yu	21st Mar
Prototype QR lookup flow	Generate sample QR code linked to a mock passport page and verify readability upon scanning.	Luxiao Cao	22nd Mar
Prepare sprint documentation	Record planning outcomes, update the report draft, and organise notes for the weekly review.	Fuyu Cao, Detai Dong	23rd Mar

4 Sprint 1 Planning (16th – 23rd March)

Table 1 comprehensively outlines the tasks, detailed descriptions, owners, and due dates for Sprint 1.

5 GitHub Repository and Jira Project Board

All group members have successfully joined the collaborative platforms required to manage this project via Scrum:

- **GitHub Repository:** https://github.com/DeTaiDong/FTGP2425_Group1_11.git
- **Jira Project Board:** https://laurencecao.atlassian.net/jira/software/projects/SCRUM/summary

EcoPassEU: A Privacy-Aware Digital Product Passport DApp for Sustainable Consumer Goods in Europe

Group Number: 11
Luxiao Cao[1], Akshansh Rajora[1], Fuyu Cao[1], Tianwei Yu[1], and Detai Dong[1]

University of Bristol, Bristol, UK
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1 What is the idea for your application?

EcoPassEU is a DApp designed to create and manage digital product passports (DPPs) for consumer goods, initially focusing on textiles and small electronics. By scanning a product-linked QR code or NFC tag, users can view a verified record of the item's provenance, material composition, repair instructions, and recycling guidance.

The core philosophy is to build a system that delivers transparency to consumers while remaining highly practical for businesses. Rather than exposing sensitive supply-chain data directly on a public ledger, the blockchain securely stores hashes, timestamps, and ownership records, whilst detailed documents are kept off-chain. This hybrid approach keeps the system cost-effective and realistic for companies guarding commercial secrets. The project aligns closely with the EU's Ecodesign for Sustainable Products Regulation (ESPR) and the broader European digital identity framework.

2 What problem(s) does it aim to solve?

EcoPassEU tackles several interconnected issues within sustainable consumption. Primarily, it combats 'greenwashing' by providing cryptographically verifiable evidence for environmental claims, allowing consumers to distinguish genuine sustainability from clever marketing.

Furthermore, as DPP requirements become mandatory across Europe, small and medium-sized enterprises (SMEs) face mounting regulatory pressure. EcoPassEU offers these businesses a low-cost, accessible tool to manage product data and demonstrate compliance without the need to build bespoke platforms. It also resolves data fragmentation across the supply chain by creating a single, tamper-evident source of truth. Ultimately, the DApp fosters trust among regulators needing auditable data, consumers seeking transparent proof, and firms requiring compliance tools that protect commercial confidentiality.

3 Who is the target audience?

The primary users are European manufacturers and brand owners—particularly SMEs in strictly regulated sectors—who require a practical solution to issue and update product compliance records.

Additionally, the DApp serves environmentally conscious consumers who want quick, reliable, and easily digestible summaries of a product's lifecycle. Market surveillance bodies and regulators form another key audience, benefiting from structured, auditable records for spot checks and investigations. In the future, the platform's utility could naturally extend to downstream stakeholders such as repair services and recyclers.

4 What techniques do you plan to use to implement the project?

At the blockchain layer, the backend will be built using Solidity smart contracts deployed on an EVM-compatible Proof-of-Stake network, such as Polygon, to ensure low gas costs and facilitate rapid iterative testing.

To represent product batches efficiently, we intend to implement the ERC-1155 semi-fungible token standard, which is better suited for batch-level tracking than minting individual NFTs for every physical item. For decentralised data storage, IPFS will host detailed documents (e.g., supplier certificates), with only the resulting content hashes committed to the blockchain. We also plan to integrate Decentralised Identifiers (DIDs) and Verifiable Credentials (VCs) to enable structured, authenticated sustainability claims, aligning with W3C standards. The frontend will be developed in React and connected via Ethers.js, prioritising a seamless user experience that abstracts Web3 complexity away from standard corporate users.

5 What is the key differentiating factor?

Unlike generic traceability prototypes that mandate radical transparency or lack practical compliance mechanisms, EcoPassEU is fundamentally grounded in privacy-aware regulatory alignment.

We recognise that enforcing absolute public disclosure of supply chain data is commercially unviable. By strictly separating on-chain cryptographic proofs from off-chain sensitive documents, we deliver verifiable sustainability without compromising corporate privacy. Furthermore, our project is deliberately scoped to be a realistic, targeted prototype rather than an overly ambitious, Europe-wide infrastructure, making it highly achievable within the constraints of this academic module.

6 What challenges or problems do you foresee?

Securing the physical-to-digital bridge remains a core challenge; while blockchain secures the digital record, physical QR codes or NFC tags can still be cloned or attached to counterfeit products.

Another critical issue is the inherent 'garbage in, garbage out' dilemma. Blockchain preserves records immutably but cannot verify the real-world truthfulness of the initial data entry, highlighting the need for robust off-chain governance. Additionally, we must carefully navigate the friction between GDPR and blockchain immutability, ensuring no sensitive data is written to the permanent ledger. Finally, achieving user adoption requires overcoming the technical intimidation associated with Web3 interfaces, necessitating a highly intuitive frontend design.

7 What questions do you have for the teaching staff?

- Is a simulated DPP workflow sufficient for this module, or is direct integration with existing external infrastructure expected?
- Is deploying to a testnet like Polygon Amoy (or a local Hardhat environment) acceptable for the prototype stage?
- How should we balance our focus between technical implementation and regulatory/conceptual alignment for grading purposes?
- Is demonstrating the physical link via a simulated QR code adequate, or should we incorporate physical NFC hardware?
- Does our proposed architecture (IPFS for documents, hashes on-chain) sufficiently address privacy and system design expectations for the final deliverable?

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FTGP Group 11: Week 22 Sprint Report

EcoPassEU — Privacy-Aware Digital Product Passport DApp

Sprint Duration: 7 April 2026 – 13 April 2026

Report Date: 13 April 2026

University of Bristol

Last Sprint Review and Retrospective

Sprint Duration: 31 March 2026 – 6 April 2026

This section summarises the sprint review meeting held on 6 April 2026 (end of Sprint 0, the initial planning and research sprint). The meeting was attended by Luxiao Cao, Akshansh Rajora, Tianwei Yu, and Detai Dong in person. Fuyu Cao participated online, having been granted leave of absence from the in-person session owing to a mid-term dissertation defence at her undergraduate institution; she nonetheless contributed to all agenda items asynchronously via the group's shared communication channel and reviewed all decisions taken before the close of the meeting.

Completed Work

The following tasks were completed during Sprint 0, together with the primary contributors responsible for each item.

Dataset identification and collection *Luxiao Cao (lead)*
Luxiao Cao took primary responsibility for sourcing and evaluating candidate open datasets suitable for populating the EcoPassEU prototype. Following a comparative review of several databases — including the Global Fashion Agenda's material database, Open Food Facts, and the ESPResso V2 LCA reference tables — the team agreed on *Open Beauty Facts* as the principal dataset for the current sprint. Luxiao Cao downloaded and decompressed the full CSV export (`en.openbeautyfacts.org.products.csv.gz`), performed an initial field audit covering all 76 available columns, and documented the dataset schema, size, and licensing terms (ODbL 1.0 / DbCL 1.0 / CC BY-SA 3.0). He identified that key fields including product name, brand, ingredient lists, packaging descriptions, country of manufacture, and sustainability certifications map directly onto the data model required by ESPR-compliant Digital Product Passports. He also flagged the ESPResso V2 reference tables as a secondary resource — particularly the `base_materials.parquet` emission factors and `aware_factors_agri.csv` water stress characterisation factors — for use in future environmental scoring modules.

Experimental design and methodology *Luxiao Cao, Akshansh Rajora, Tianwei Yu; Fuyu Cao (online)*
Drawing upon the ESPResso V2 LCA framework and ISO 14040/14044 methodology, the team collectively formulated the experimental pipeline for the EcoPassEU prototype. The agreed design proceeds in three stages: (1) ingestion and normalisation of Open Beauty Facts records, including country name standardisation and imputation of missing packaging fields using documented fallback values; (2) serialisation of normalised records to structured JSON-LD documents, off-chain storage on IPFS, and on-chain commitment

of content hashes via an ERC-1155 smart contract on the Polygon Amoy testnet; and (3) functional and non-functional evaluation covering hash integrity verification, QR code resolution, gas cost per DPP issuance, and frontend load time. Fuyu Cao participated in the design discussion online, contributing written comments on the LCA methodology alignment and data gap handling strategy. The methodology was written up and submitted as the group's first weekly progress report in Springer LNCS format.

System architecture design *Akshansh Rajora, Tianwei Yu*
Akshansh Rajora and Tianwei Yu produced the first draft of the system architecture diagram, specifying the interaction between the React frontend, the Ethers.js middleware layer, the Solidity smart contracts, and the IPFS off-chain storage layer. The draft was reviewed by the full group and committed to the repository under `docs/architecture_v1.pdf`.

Smart contract skeleton and development environment *Detai Dong*
Detai Dong initialised the Hardhat development environment and authored a skeleton Solidity smart contract implementing the ERC-1155 semi-fungible token standard. The contract exposes stub functions for `mintBatch`, `updatePassportHash`, and `getPassportURI`, and compiles without errors under Solidity 0.8.24. Unit test scaffolding using Chai and Hardhat's built-in network was also put in place. Detai Dong additionally began planning the React frontend component architecture, identifying the key views required for passport display, QR code resolution, and wallet connection.

Changes and Incomplete Tasks

Data cleaning pipeline — deferred to Sprint 1. The normalisation and cleaning pipeline for the Open Beauty Facts CSV was planned for Sprint 0 but was not completed. The raw file is larger than initially estimated (approximately 2.8 GB decompressed), and the team required additional time to agree on a filtering strategy to produce a manageable and representative working subset. This task has been carried forward as a high-priority item for Sprint 1.

Decentralised Identifier (DID) integration — scope reduced. The original proposal included W3C Decentralised Identifiers (DIDs) and Verifiable Credentials (VCs) as first-class features of the smart contract layer. After architectural review, the team agreed to defer full DID/VC integration to a later sprint and instead simulate the credential structure using JSON-LD documents stored on IPFS. This change reduces scope risk without compromising the conceptual alignment with the ESPR framework.

Fuyu Cao's contributions — adjusted delivery mode. Due to Fuyu Cao's midterm dissertation defence at her undergraduate institution, her planned contributions to the data pipeline were delivered asynchronously this sprint. She participated in all team discussions online, reviewed all technical decisions, and submitted her written contributions (regulatory notes, dataset evaluation comments) via the group's shared repository. Her role in Sprint 1 will be adjusted to tasks that are better suited to asynchronous collaboration, whilst she remains available online throughout.

Retrospective

Start Doing	Stop Doing	Keep Doing
Set explicit file-size and format constraints when identifying datasets, to avoid late-stage pipeline surprises.	Underestimating the volume of open datasets; always perform a quick field audit before committing to a source.	Daily asynchronous standups via the group chat, which have kept everyone informed without requiring synchronous attendance.
Use feature branches in Git for each sprint task, with pull requests reviewed by at least one other team member before merging.	Leaving architecture decisions undocumented until the review meeting; agree on and record decisions in real time.	Distributing written contributions (notes, drafts) to the shared repository at least 24 hours before meetings to allow async review.
Allocate a dedicated slot in each sprint for integration testing, rather than treating it as an afterthought at the end of development.	Scope creep in technical tasks: DID/VC integration was over-specified for this stage and needed to be deferred.	Inclusive participation: ensuring members attending remotely (as Fuyu Cao did this sprint) have equal access to agenda items and decision records.

Next Sprint Planning

Sprint Duration: 14 April 2026 – 20 April 2026

The sprint planning meeting for Sprint 1 was held on 13 April 2026. The full team was present; Fuyu Cao attended online. The sprint vision, role assignments, and backlog are detailed below.

Product Owner

Akshansh Rajora

Scrum Master

Tianwei Yu

Sprint Vision

By the end of Sprint 1, the team aims to have a working end-to-end prototype connecting the data ingestion pipeline to a functional user-facing interface. A cleaned subset of Open Beauty Facts records should flow through a normalisation script, be committed to IPFS, and have their content hashes stored in the deployed ERC-1155 smart contract on the Polygon Amoy testnet. Crucially, a fully implemented React frontend — covering passport display, QR code resolution, and wallet connection — should be delivered and ready for integration testing by the close of the sprint.

Sprint Backlog

The following tasks have been selected from the product backlog for Sprint 1. Estimates are given in story points (1 point ≈ half a day of focused work).

Task	**Description**	**Assignee(s)**	**Points**
Data cleaning and normalisation	Filter Open Beauty Facts CSV to ∼500 records, resolve encoding issues, standardise country names, and impute missing fields with documented fallback values.	Luxiao Cao (lead), Fuyu Cao (async)	5
Data gap documentation	Produce a structured CSV log of all imputed fields, recording original value, substituted value, and justification.	Fuyu Cao (async)	2
IPFS integration and smart contract	Serialise records to JSON-LD, pin to IPFS, and extend the ERC-1155 contract to store CIDs per token with unit tests.	Tianwei Yu, Akshansh Rajora	6
Testnet deployment	Deploy the smart contract to Polygon Amoy via Hardhat Ignition; commit contract address and ABI to the repository.	Tianwei Yu	2
Frontend: full implementation	Build the complete React frontend: passport display page, QR code scanner/generator, and Ethers.js wallet connection for DPP issuance.	Detai Dong	8
Integration test	End-to-end flow test: ingest → IPFS → mint → QR → display. Record pass/fail, gas costs, and load time.	All members	3

Total story points this sprint: 26

Notes on Member Availability

Fuyu Cao remains on leave of absence from in-person sessions for the duration of Sprint 1 due to her undergraduate mid-term dissertation defence obligations. She will continue to participate online in all planning, review, and retrospective meetings, and will take ownership of the data gap documentation task, which is well-suited to asynchronous delivery. All other team members are available full-time in Bristol for the sprint duration.

Detai Dong has committed to delivering the complete React frontend by the end of Sprint 1. Given the scope of the frontend task (8 story points), this is the highest individual workload in the sprint; the team has agreed to prioritise frontend integration support in the final two days of the sprint to ensure the end-to-end flow test can proceed on schedule.

Anything Else to Share

The team has opened a GitHub repository under a private organisation account for version control and documentation. All sprint artefacts completed to date — including the architecture diagram and dataset schema notes — are committed there. Access has been shared with the supervising teaching staff as requested.

We have also identified a potential risk regarding the IPFS pinning strategy: free-tier pinning services such as Pinata impose monthly bandwidth and storage caps that may become restrictive if the dataset subset grows beyond the planned 500 records. The team will monitor usage during Sprint 1 and evaluate whether a self-hosted IPFS node on a university virtual machine would be more appropriate for the prototype evaluation phase.

Finally, the team notes that the choice of Open Beauty Facts as the primary dataset was confirmed as appropriate during internal review this week. The dataset's broad field coverage (ingredients, packaging, provenance, certifications) maps well onto the DPP data model, and its open licensing allows us to publish example passport records in the repository without restriction. Textile-specific data integration, drawing on the ESPResso V2 carbon and water footprint tables, remains planned for a later sprint.

EcoPassEU — Group 11 | University of Bristol | Sprint Report 1 | 13 April 2026

FTGP Group 11: Week 23 Sprint Report

EcoPassEU — Privacy-Aware Digital Product Passport DApp

Sprint Duration: 120 April 2026 – 30 April 2026

Report Date: 20 April 2026

University of Bristol

Last Sprint Review and Retrospective

Sprint Duration: 14 April 2026 – 20 April 2026

This section summarises the sprint review meeting held on 20 April 2026, marking the close of Sprint 1. The meeting was attended by all five team members in person: Luxiao Cao, Akshansh Rajora, Tianwei Yu, Detai Dong, and Fuyu Cao, who returned to in-person attendance this sprint having fulfilled her undergraduate mid-term dissertation defence obligations.

Completed Work

The following tasks were completed during Sprint 1, together with the primary contributors responsible for each item.

All sprint artefacts committed to GitHub *All members*
All work produced during Sprint 1 was committed to the group's private GitHub repository under the organisation account shared with supervising teaching staff. This includes source code, data-processing scripts, smart contract code, and documentation. The repository is accessible at `https://github.com/DeTaiDong/FTGP2425_Group1_11`.

Paper writing: introduction and background *Luxiao Cao (lead)*
Luxiao Cao led the academic writing effort this sprint, establishing the structural framework for the project paper and authoring both the introduction and background sections. The introduction situates EcoPassEU within the European Union's Ecodesign for Sustainable Products Regulation (ESPR) and articulates the motivation for a privacy-preserving, blockchain-based Digital Product Passport. The background section surveys relevant prior work spanning Life Cycle Assessment (LCA) methodology, decentralised identifier frameworks, and IPFS-based off-chain storage architectures. Both sections were written in Springer LNCS format, consistent with the group's agreed submission template.

Development environment support and application testing *Luxiao Cao*
In addition to his writing responsibilities, Luxiao Cao assisted other team members in configuring and troubleshooting their local development environments, in particular resolving Node.js version conflicts encountered when running the Hardhat toolchain alongside the React frontend scaffold. He also conducted an initial round of functional testing on the deployed application, verifying wallet connection, QR code generation, and passport retrieval flows, and documented the results in a shared testing log committed to the repository.

Frontend: core implementation *Detai Dong (lead)*
Detai Dong substantially completed the React frontend during Sprint 1. The delivered

implementation covers the three principal views specified in the sprint backlog: the passport display page, which renders normalised DPP data retrieved from IPFS; the QR code scanner and generator, integrated with the `qrcode.react` library; and the wallet connection component, implemented via Ethers.js and tested against the Polygon Amoy testnet. Detai Dong notes that a small number of supplementary features — including export-to-PDF and a comparative product view — were scoped but not yet delivered; these are planned for a subsequent sprint. The core functionality required for end-to-end integration testing is in place.

Smart contract and partial backend *Akshansh Rajora*
Akshansh Rajora completed the ERC-1155 smart contract implementation, extending the skeleton from Sprint 0 to support full IPFS Content Identifier (CID) storage per token, on-chain event emission for passport issuance and updates, and access-controlled `mintBatch` and `updatePassportHash` functions. The contract was deployed to the Polygon Amoy testnet via Hardhat Ignition; the contract address and ABI were committed to the repository under `contracts/deployments/`. Akshansh Rajora additionally implemented supporting backend utility functions for CID retrieval and event indexing, which are consumed by the React frontend via Ethers.js.

Data processing and contract-ready field mapping *Fuyu Cao (async), Tianwei Yu*
Fuyu Cao and Tianwei Yu jointly produced the data pipeline for Sprint 1. Working from the Open Beauty Facts CSV downloaded in Sprint 0, they applied a filtering strategy to produce a representative working subset of approximately 500 records, resolved UTF-8 encoding issues in the source file, standardised country-of-manufacture names to ISO 3166-1 alpha-2 codes, and imputed missing packaging fields using the documented fallback values agreed in Sprint 0. Critically, and in response to the schema-alignment issue identified during integration, the team ensured that all output JSON-LD field names and data types conform precisely to the smart contract ABI. Fuyu Cao produced the data gap log — a structured CSV recording each imputed field, its original value, the substituted value, and the justification — and submitted it to the repository ahead of the review meeting.

Changes and Incomplete Tasks

Frontend supplementary features — deferred to Sprint 2.
Detai Dong's additional frontend features (export-to-PDF, comparative product view) were not completed within the sprint. These are lower-priority enhancements relative to the core integration deliverables and have been carried forward to the Sprint 2 backlog.

Schema alignment issue — resolved mid-sprint.
An unanticipated mismatch was identified between the field names produced by the initial data pipeline and those expected by the smart contract ABI. This required a mid-sprint realignment between Fuyu Cao and Tianwei Yu (data pipeline) and Akshansh Rajora (smart contract), adding approximately half a day of unplanned work. The issue was resolved before the integration test and highlighted the need for an explicit shared schema document at the outset of future sprints.

Retrospective

Start Doing	Stop Doing	Keep Doing
Establish a formal mid-sprint check-in to review integration progress and surface blockers before the final two days of the sprint.	Treating integration testing as a final-day activity; begin smoke-testing component interfaces as soon as individual tasks reach a stable state.	Daily asynchronous standups, which continued to keep all members informed across time zones without requiring synchronous attendance.
Define and commit shared data-contract schemas (JSON-LD field names, types, and nullable flags) at the outset of each sprint, before implementation begins.	Leaving field naming conventions implicit; schema disagreements between the data pipeline and the smart contract cost time during integration.	Feature branches with pull-request review; the practice was adopted this sprint and reduced merge conflicts noticeably.
Include a brief written risk update in the repository at each sprint mid-point, flagging any emerging technical risks (e.g. IPFS bandwidth, testnet faucet availability).	Deferring risk documentation until the retrospective; risks identified late in the sprint (IPFS pinning limits) would benefit from earlier visibility.	Inclusive participation: all five team members contributed actively throughout the sprint, with clear task ownership and timely communication maintaining momentum across the group.

Next Sprint Planning

Sprint Duration: 21 April 2026 – 27 April 2026

The sprint planning meeting for Sprint 2 was held on 20 April 2026. The full team was present in person.

Product Owner

Tianwei Yu

Scrum Master

Detai Dong

Sprint Vision

By the end of Sprint 2, the team aims to have a fully integrated and evaluated prototype ready for academic reporting. The principal objectives are: completing the remaining frontend features; expanding and refining the academic paper with methodology, results, and evaluation sections; conducting a comprehensive evaluation covering hash integrity, gas costs, QR resolution latency, and frontend load time; and addressing any outstanding technical debt from Sprint 1. The sprint will conclude with a near-final draft of the project paper and a stable, demonstrable prototype.

Sprint Backlog

The following tasks have been selected from the product backlog for Sprint 2. Estimates are given in story points (1 point ≈ half a day of focused work).

Task	Description	Assignee(s)	Points
Paper writing: methodology, results & evaluation	Draft the methodology, results, and evaluation sections of the project paper, incorporating gas cost data, load time measurements, and hash integrity results from Sprint 1.	Luxiao Cao (lead)	5
Application testing & bug fixes	Conduct systematic regression testing of the full application stack; document and resolve outstanding bugs identified during Sprint 1 integration testing.	Luxiao Cao	2
Frontend: supplementary features	Implement the deferred frontend features: export-to-PDF for passport records and a comparative product view for side-by-side DPP display.	Detai Dong (lead)	4
Smart contract: audit & optimisation	Review the deployed ERC-1155 contract for gas optimisation opportunities; add NatSpec documentation; update unit tests to achieve full branch coverage.	Akshansh Rajora	4
Data pipeline: extended dataset & ESPResso integration	Expand the working dataset to 1,000 records and integrate ESPResso V2 emission factors and water stress characterisation factors into the JSON-LD output.	Fuyu Cao, Tianwei Yu	5
Comprehensive evaluation	Run the full evaluation suite: hash integrity verification, QR code resolution latency (10 samples), gas cost per DPP issuance (mean and variance), and frontend load time. Produce tables and figures for the paper.	All members	5

Total story points this sprint: 25

Notes on Member Availability

Fuyu Cao has returned to in-person attendance from Sprint 2 onwards, having completed her undergraduate mid-term dissertation defence obligations. She will attend all sprint meetings in person alongside the rest of the team and will take ownership of the data pipeline expansion and ESPResso integration task.

All other team members are available full-time in Bristol for the sprint duration. Given the volume of academic writing required this sprint, Luxiao Cao has indicated that writing productivity may be sensitive to interruptions; the team has agreed to protect at least two uninterrupted working days for paper drafting. Detai Dong has confirmed that the deferred frontend features are scoped to four story points and are achievable within the sprint without impacting integration stability.

Anything Else to Share

The GitHub repository continues to serve as the single source of truth for all sprint artefacts. All code, documentation, data pipeline scripts, and test logs from Sprint 1 are committed and accessible to supervising teaching staff at https://github.com/DeTaiDong/FTGP2425_Group1_11.

The IPFS pinning bandwidth concern flagged in the Week 22 report remains under monitoring. Usage during Sprint 1, with a dataset subset of approximately 500 records, remained within the free-tier limits of the Pinata pinning service. Should the dataset grow significantly in Sprint 2 — as planned, to 1,000 records — the team will evaluate deployment of a self-hosted IPFS node on a university virtual machine as a contingency.

The end-to-end integration test conducted at the close of Sprint 1 was successful: all 500 records were ingested, pinned to IPFS, minted as ERC-1155 tokens on Polygon Amoy, and resolved correctly via QR code in the React frontend. Gas costs per DPP issuance and frontend load times were within acceptable parameters and will be reported formally in the Sprint 2 evaluation section of the paper.

EcoPassEU — Group 11 | University of Bristol | Sprint Report W23 | 20 April 2026

FTGP Group 11: Week 24 Sprint Report

EcoPassEU — Privacy-Aware Digital Product Passport DApp

Sprint Duration: 28 April 2026 – 4 May 2026

Report Date: 30 April 2026

University of Bristol

Last Sprint Review and Retrospective

Sprint Duration: 21 April 2026 – 27 April 2026

This section summarises the sprint review meeting held on 28 April 2026, marking the close of Sprint 2. The meeting was attended by all five team members in person: Luxiao Cao, Akshansh Rajora, Tianwei Yu, Detai Dong, and Fuyu Cao.

Completed Work

The following tasks were completed during Sprint 2, together with the primary contributors responsible for each item.

All sprint artefacts committed to GitHub *All members*

All work produced during Sprint 2 was committed to the group's GitHub repository under the organisation account shared with supervising teaching staff. This includes revised paper drafts, updated smart contract code and ABI files, frontend deployment assets, Hardhat test outputs, and expanded data pipeline scripts. The repository is accessible at `https://github.com/DeTaiDong/FTGP2425_Group1_11`.

Paper writing: introduction, background, and design documentation *Luxiao Cao (lead)*

Luxiao Cao continued his leading role in academic writing this sprint, completing and refining the introduction and background sections carried over from Sprint 1, and producing a full design documentation chapter covering the system architecture of EcoPassEU. The design documentation details the off-chain storage model using IPFS and the Pinata pinning service, the on-chain ERC-1155 token structure and its relationship to Content Identifier (CID) storage, the JSON-LD schema used for Digital Product Passport records, and the interaction model between the React frontend and the Polygon Amoy testnet via Ethers.js. Luxiao Cao additionally conducted a systematic literature survey, identifying and integrating references across blockchain-based provenance systems, the EU Ecodesign for Sustainable Products Regulation (ESPR) framework, and Life Cycle Assessment (LCA) methodologies. All citations were formatted in compliance with the Springer LNCS style agreed by the group.

Hardhat testing *Luxiao Cao*

In addition to his paper writing responsibilities, Luxiao Cao authored and executed a Hardhat test suite covering the deployed ERC-1155 smart contract. The suite tests all principal contract entry points—`mintBatch`, `updatePassportHash`, and the CID retrieval utilities—verifying correct event emission, access-control enforcement, reversion on invalid inputs, and gas consumption under representative call patterns. All tests passed against the revised contract deployment. The results were committed to the repository and will serve as a primary data source for the gas cost analysis reported in the evaluation section of the paper.

Frontend: deployment and supplementary features *Detai Dong (lead)*

Detai Dong completed the production deployment of the React frontend and delivered both supplementary features deferred from Sprint 1. The application was deployed via Vercel, with environment variables for the Polygon Amoy RPC endpoint and the deployed contract address injected at build time; a `vercel.json` configuration was committed to the repository to document the deployment parameters. The supplementary features delivered this sprint are: an export-to-PDF function for individual passport records, implemented using a client-side rendering approach to preserve layout fidelity without server-side dependencies; and a comparative product view enabling users to display two Digital Product Passport records side by side for direct comparison. Both features were integrated into the existing routing structure and verified against the live deployment without regressions to the core passport retrieval or QR code flows.

Smart contract: audit and optimisation *Akshansh Rajora*

Akshansh Rajora conducted a comprehensive review of the deployed ERC-1155 contract, identifying and implementing several gas optimisation opportunities. Changes applied include elimination of redundant storage reads within view functions, consolidation of emitted event parameters to reduce per-transaction log data volume, and tightening of access-control modifiers to remove unnecessary conditional branching. NatSpec documentation was added to all public and external functions in accordance with Ethereum development conventions. The unit test suite was updated to achieve full branch coverage. The revised contract was redeployed to the Polygon Amoy testnet; the updated contract address and ABI were committed to `contracts/deployments/`.

Data pipeline: extended dataset and ESPResso integration *Fuyu Cao (lead), Tianwei Yu*

Fuyu Cao and Tianwei Yu expanded the working dataset from approximately 500 to 1,000 records, applying the filtering and normalisation pipeline established in Sprint 1 to the additional records and resolving a small number of previously unencountered UTF-8 encoding variants in the source CSV. Fuyu Cao led the ESPResso V2 integration effort, researching the characterisation factor tables, mapping emission factor identifiers to the existing product categories in the Open Beauty Facts dataset, and implementing the field population logic in the data pipeline script. Each passport record in the output now carries `co2Equivalent` and `waterStressIndex` fields populated from the ESPResso tables. Tianwei Yu took responsibility for the batch minting pipeline, coordinating the upload of expanded JSON-LD records to IPFS via Pinata and the subsequent minting of ERC-1155 tokens on Polygon Amoy, verifying that all 1,000 CIDs were correctly stored on-chain and resolvable. Total pinning bandwidth usage remained within the free-tier limits of the Pinata service. Fuyu Cao updated the data gap log to document all imputation decisions made during the expansion, including the justification for fallback values applied to records with missing packaging or ingredient fields.

Sprint coordination and product backlog management *Tianwei Yu*

In his capacity as Product Owner for Sprint 2, Tianwei Yu maintained and prioritised the product backlog throughout the sprint, facilitating daily asynchronous standups and ensuring that dependencies between the data pipeline, smart contract, and frontend workstreams were surfaced and resolved promptly. He coordinated the mid-sprint realignment required when the ESPResso field mapping introduced a minor schema update, confirming with Akshansh Rajora that the on-chain ABI remained compatible and that no contract redeployment was required at that stage. Tianwei Yu additionally prepared the integration test plan used at the close of the sprint, specifying the verification criteria for the full 1,000-record end-to-end flow, and ran the integration test alongside Fuyu Cao to confirm successful resolution of all minted tokens via QR code in the deployed frontend.

Preliminary evaluation data collection *Fuyu Cao, Tianwei Yu*

Fuyu Cao and Tianwei Yu initiated the evaluation suite in the final days of the sprint, collecting a preliminary set of measurements ahead of the formal evaluation planned for Sprint 3. Fuyu Cao ran hash integrity verification across a representative sample of 200 records from the expanded dataset, confirming that the SHA-256 digest computed client-side from IPFS-retrieved JSON-LD matched the value stored on-chain for every record in the sample. Tianwei Yu recorded ten QR code resolution latency samples under typical campus network conditions, capturing the time from QR code scan to full passport data render in the frontend. These preliminary results were committed to the repository as a structured CSV and will be extended to the full 1,000-record dataset in Sprint 3.

Changes and Incomplete Tasks

Methodology, results, and evaluation sections — deferred to Sprint 3.

The paper sections covering methodology, quantitative results, and formal evaluation were not completed within Sprint 2. Whilst gas cost data and preliminary QR latency and hash integrity measurements were collected, the full evaluation suite was not finalised in time to support complete paper drafting. These sections constitute the critical-path deliverable for Sprint 3.

Comprehensive evaluation suite — partially completed.

Gas cost measurements (via Hardhat), a preliminary hash integrity sample (200 records, Fuyu Cao), and ten QR resolution latency samples (Tianwei Yu) were collected during the sprint. Full hash integrity verification across all 1,000 records and frontend load time profiling remain outstanding and will be completed as the first priority of Sprint 3.

Retrospective

Start Doing	Stop Doing	Keep Doing
Draft results and evaluation sections incrementally as measurement data becomes available, rather than waiting for the complete dataset before beginning to write.	Treating paper writing as a strictly sequential activity that must follow all engineering work; writing and evaluation can proceed concurrently once component interfaces are stable.	Daily asynchronous standups, which continued to surface blockers promptly and keep all members aligned across parallel workstreams throughout the sprint.
Allocate a time-boxed evaluation window early in the sprint so that measurements are available to inform paper drafts throughout the sprint rather than only at its close.	Deferring evaluation planning until individual components reach full stability; smoke-level measurements can commence as soon as a component is in a testable state.	Feature branch discipline and pull-request review, which continued to provide a lightweight quality gate for both codebase and paper draft changes, and reduced merge conflicts noticeably.
Prepare a formal submission checklist at the outset of Sprint 3, covering code freeze, paper formatting compliance, repository archival, and submission portal requirements.		Clear task ownership and proactive cross-team communication, which maintained sprint velocity despite the breadth of simultaneous engineering, writing, and evaluation workstreams.

Next Sprint Planning

Sprint Duration: 28 April 2026 – 4 May 2026

The sprint planning meeting for Sprint 3 was held on 28 April 2026. The full team was present in person.

Product Owner

Tianwei Yu

Scrum Master

Detai Dong

Sprint Vision

Sprint 3 is the final sprint of the project. By the close of this sprint, the team aims to deliver a complete, submission-ready project paper and a fully evaluated, stable prototype. The principal deliverables are: the methodology, results, and evaluation sections of the paper; the full evaluation suite executed across the 1,000-record dataset; a finalised and proofread document compiled in Springer LNCS format; and a confirmed production deployment with the repository in an archival state accessible to the examiners. No new features will be added to the prototype this sprint; all engineering effort is directed at evaluation, documentation, and submission preparation.

Sprint Backlog

The following tasks have been selected from the product backlog for Sprint 3. Estimates are given in story points (1 point ≈ half a day of focused work).

Task	Description	Assignee(s)	Points
Paper writing: methodology, results & evaluation	Draft and finalise the methodology, results, and evaluation sections of the project paper, incorporating gas cost data, QR latency measurements, frontend load time figures, and hash integrity results. Produce all required tables and figures in Springer LNCS format.	Luxiao Cao (lead)	5
Comprehensive evaluation suite	Complete the full evaluation suite: hash integrity verification across all 1,000 records, QR code resolution latency (10 samples), gas cost per DPP issuance (mean and variance over 20 transactions), and frontend load time under representative network conditions. Extend preliminary data collected in Sprint 2 to the full dataset.	Fuyu Cao, Tianwei Yu (lead), all members	4
Conclusion and abstract	Write the paper conclusion synthesising the principal findings from the evaluation; revise the abstract to reflect the final scope and results of the project.	Luxiao Cao (lead)	2
Paper finalisation and proofreading	Integrate all sections into a single compiled document in Springer LNCS format; resolve cross-reference, citation, and formatting inconsistencies; conduct a full proofreading pass to ensure consistent British English spelling and academic register throughout.	Luxiao Cao, Fuyu Cao	3
Final deployment and repository archival	Verify the production deployment resolves all 1,000 records correctly; confirm all contract addresses, ABIs, and environment configuration files are current; archive the repository in the state required by the examiners.	Detai Dong, Akshansh Rajora	2
		Total story points this sprint:	**16**

Notes on Member Availability

All five team members are available full-time in Bristol for the duration of Sprint 3. Given that this is the final sprint and the submission deadline is imminent, the team has agreed that all engineering effort will be focused exclusively on evaluation and deployment verification; no further feature development will be undertaken. Luxiao Cao has reserved the final two days of the sprint for paper compilation, formatting, and proofreading. Fuyu Cao will assist with proofreading and reference verification in addition to leading the remaining hash integrity evaluation runs. Tianwei Yu will coordinate the full evaluation suite as Product Owner and will lead the QR latency and load time measurement sessions. Detai Dong and Akshansh Rajora will be available throughout the sprint to support any evaluation measurements requiring live application access.

Anything Else to Share

The GitHub repository continues to serve as the single source of truth for all project artefacts. All code, data pipeline scripts, test outputs, deployment configurations, and paper drafts from Sprints 1 and 2 are committed and accessible to supervising teaching staff at https://github.com/DeTaiDong/FTGP2425_Group1_11.

IPFS pinning usage following the expansion to 1,000 records remains within the Pinata free-tier bandwidth limits. Deployment of a self-hosted IPFS node is no longer considered necessary for the remainder of the project.

The end-to-end integration test conducted at the close of Sprint 2 by Fuyu Cao and Tianwei Yu confirmed that all 1,000 records are correctly pinned to IPFS, minted as ERC-1155 tokens on Polygon Amoy, and resolvable via QR code in the deployed React frontend. The prototype is stable and meets the requirements for the final evaluation and paper submission.

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A.10 Smart Contract and Test Codes

Listing 1.6. Complete PassportRegistry Smart Contract Code

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract PassportRegistry {

    struct Passport {
        string  ipfsCID;       // off-chain document pointer
        bytes32 metadataHash;  // sha256 of the JSON for tamper-proof verify
        address issuer;
        uint256 timestamp;
        bool    exists;
    }

    // productId => Passport
    mapping(string => Passport) private passports;

    event PassportIssued(
    string indexed productId,
    string ipfsCID,
    bytes32 metadataHash,
    address indexed issuer,
    uint256 timestamp
    );

    // Single register
    function registerPassport(
    string calldata productId,
    string calldata ipfsCID,
    bytes32 metadataHash
    ) external {
        require(!passports[productId].exists, "Passport already exists");
        passports[productId] = Passport({
            ipfsCID:      ipfsCID,
            metadataHash: metadataHash,
            issuer:       msg.sender,
            timestamp:    block.timestamp,
            exists:       true
        });
        emit PassportIssued(productId, ipfsCID, metadataHash, msg.sender, block.timestamp);
    }

```

```
    // Batch register (gas efficient for SMEs uploading product ranges)
    function batchRegisterPassports(
    string[]   calldata productIds,
    string[]   calldata ipfsCIDs,
    bytes32[]  calldata metadataHashes
    ) external {
        require(
        productIds.length == ipfsCIDs.length &&
        ipfsCIDs.length == metadataHashes.length,
        "Array length mismatch"
        );
        for (uint256 i = 0; i < productIds.length; i++) {
            if (!passports[productIds[i]].exists) {
                passports[productIds[i]] = Passport({
                    ipfsCID:      ipfsCIDs[i],
                    metadataHash: metadataHashes[i],
                    issuer:       msg.sender,
                    timestamp:    block.timestamp,
                    exists:       true
                });
                emit PassportIssued(productIds[i], ipfsCIDs[i], metadataHashes[i], msg.sender, block.timestamp);
            }
        }
    }

    // Public read - anyone can verify a product
    function getPassport(string calldata productId)
    external view
    returns (string memory ipfsCID, bytes32 metadataHash, address issuer, uint256 timestamp)
    {
        Passport storage p = passports[productId];
        require(p.exists, "Passport not found");
        return (p.ipfsCID, p.metadataHash, p.issuer, p.timestamp);
    }

    function passportExists(string calldata productId) external view returns (bool) {
        return passports[productId].exists;
    }
}
```

Listing 1.7. Hardhat unit testing suite for PassportRegistry.sol

```
import { expect } from "chai";
import hre from "hardhat";

describe("PassportRegistry Smart Contract",
    function () {
        let contract;
        let owner;
        let addr1;

        const PRODUCT_ID = "ECO-TX-2025-001";
        const IPFS_CID   = "
            QmTestCID123456789";
        const META_HASH  = hre.ethers.
            keccak256(hre.ethers.toUtf8Bytes
            ("test-metadata"));

        beforeEach(async function () {
                [owner, addr1] = await hre.
                    ethers.getSigners();
                const Factory = await hre.
                    ethers.getContractFactory
                    ("PassportRegistry");
                contract = await Factory.
                    deploy();
        });

        // 1. Deployment
        describe("Deployment", function () {
                it("should deploy with a
                    valid contract address",
                    async function () {
                        expect(contract.
                            target).to.be.a("
                            string");
                        expect(contract.
                            target).to.match
                            (/^0x[0-9a-fA-F
                            ]{40}$/);
                });
        });

        // 2. registerPassport
        describe("registerPassport", function
             () {
                it("should register a new
                    passport successfully",
                    async function () {
                        await expect(
                        contract.
                            registerPassport(
```

```
                PRODUCT_ID,
                IPFS_CID,
                META_HASH)
        ).to.emit(contract, "
                PassportIssued");
});

it("should store the correct
    issuer address", async
    function () {
        await contract.
            registerPassport(
            PRODUCT_ID,
            IPFS_CID,
            META_HASH);
        const [, , issuer] =
            await contract.
            getPassport(
            PRODUCT_ID);
        expect(issuer).to.
            equal(owner.
            address);
});

it("should store the correct
    IPFS CID and metadata
    hash", async function ()
    {
        await contract.
            registerPassport(
            PRODUCT_ID,
            IPFS_CID,
            META_HASH);
        const [cid, hash] =
            await contract.
            getPassport(
            PRODUCT_ID);
        expect(cid).to.equal(
            IPFS_CID);
        expect(hash).to.equal
            (META_HASH);
});

it("should record a non-zero
    timestamp", async
    function () {
        await contract.
            registerPassport(
            PRODUCT_ID,
```

```
                        IPFS_CID,
                        META_HASH);
                    const [, , ,
                        timestamp] =
                        await contract.
                        getPassport(
                        PRODUCT_ID);
                    expect(Number(
                        timestamp)).to.be
                        .greaterThan(0);
            });

            it("should revert when
                registering a duplicate
                product ID", async
                function () {
                    await contract.
                        registerPassport(
                        PRODUCT_ID,
                        IPFS_CID,
                        META_HASH);
                    await expect(
                    contract.
                        registerPassport(
                        PRODUCT_ID,
                        IPFS_CID,
                        META_HASH)
                    ).to.be.revertedWith
                        ("Passport
                        already exists");
            });
    });

    // 3. passportExists
    describe("passportExists", function
        () {
            it("should return false for
                an unregistered product",
                 async function () {
                    expect(await contract
                        .passportExists("
                        NONEXISTENT-ID"))
                        .to.equal(false);
            });

            it("should return true after
                registration", async
                function () {
                    await contract.
                        registerPassport(
```

```
                        PRODUCT_ID,
                        IPFS_CID,
                        META_HASH);
                    expect(await contract
                        .passportExists(
                        PRODUCT_ID)).to.
                        equal(true);
            });
    });

    // 4. getPassport
    describe("getPassport", function () {
            it("should revert when
                querying a non-existent
                product", async function
                () {
                    await expect(
                    contract.getPassport
                        ("NO-SUCH-PRODUCT
                        ")
                    ).to.be.revertedWith
                        ("Passport not
                        found");
            });

            it("should return all four
                fields correctly", async
                function () {
                    await contract.
                        registerPassport(
                        PRODUCT_ID,
                        IPFS_CID,
                        META_HASH);
                    const [cid, hash,
                        issuer, timestamp
                        ] = await
                        contract.
                        getPassport(
                        PRODUCT_ID);
                    expect(cid).to.equal(
                        IPFS_CID);
                    expect(hash).to.equal
                        (META_HASH);
                    expect(issuer).to.
                        equal(owner.
                        address);
                    expect(Number(
                        timestamp)).to.be
                        .greaterThan(0);
            });
```

```
});

// 5. batchRegisterPassports
describe("batchRegisterPassports",
  function () {
    const ids    = ["BATCH-001",
      "BATCH-002", "BATCH
      -003"];
    const cids   = ["QmCID1", "
      QmCID2", "QmCID3"];
    const hashes = [
    hre.ethers.keccak256(hre.
      ethers.toUtf8Bytes("hash1
      ")),
    hre.ethers.keccak256(hre.
      ethers.toUtf8Bytes("hash2
      ")),
    hre.ethers.keccak256(hre.
      ethers.toUtf8Bytes("hash3
      ")),
    ];

    it("should register multiple
      products in one
      transaction", async
      function () {
        await contract.
          batchRegisterPassports
          (ids, cids,
          hashes);
        for (const id of ids)
          {
            expect(await
              contract.
              passportExists
              (id)).to.
              equal(
              true);
        }
    });

    it("should emit
      PassportIssued for each
      new product", async
      function () {
        const tx = await
          contract.
          batchRegisterPassports
          (ids, cids,
          hashes);
```

```
        const receipt = await
             tx.wait();
        const events =
            receipt.logs.
            filter(
        log => log.fragment
            && log.fragment.
            name === "
            PassportIssued"
        );
        expect(events.length)
            .to.equal(ids.
            length);
});

it("should revert when array
    lengths do not match",
    async function () {
        await expect(
        contract.
            batchRegisterPassports
            (["ID-1"], ["CID
            -1", "CID-2"], [
            hashes[0]])
        ).to.be.revertedWith
            ("Array length
            mismatch");
});

it("should silently skip
    already-registered
    products in a batch",
    async function () {
        await contract.
            registerPassport(
            ids[0], cids[0],
            hashes[0]);
        await expect(
        contract.
            batchRegisterPassports
            (ids, cids,
            hashes)
        ).to.not.be.reverted;
        expect(await contract
            .passportExists(
            ids[1])).to.equal
            (true);
        expect(await contract
            .passportExists(
```

```
                                ids[2])).to.equal
                                (true);
                    });
                });

                // 6. PassportIssued event
                describe("PassportIssued event",
                    function () {
                        it("should emit correct
                            values on single
                            registration", async
                            function () {
                                await expect(
                                contract.
                                    registerPassport(
                                    PRODUCT_ID,
                                    IPFS_CID,
                                    META_HASH)
                                )
                                .to.emit(contract, "
                                    PassportIssued")
                                .withArgs(
                                PRODUCT_ID,
                                IPFS_CID,
                                META_HASH,
                                owner.address,
                                (ts) => Number(ts) >
                                    0
                                );
                    });
                });
            });
```

Frontend Core Files Due to the extensive nature of the React frontend architecture, the complete suite of source files—including all core components, modifications, and user interface logic—is hosted on our group's GitHub repository.

The full frontend implementation can be reviewed at the following link:

– `https://github.com/DeTaiDong/FTGP2526_Group1_11`